

Jorge Ramirez

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Education

California Polytechnic State University

San Luis Obispo, CA

M.S. in Electrical Engineering

Sep. 2025 – Dec 2026

B.S. in Computer Engineering

Sep. 2021 – June 2025

Relevant Coursework: Vision Based Robotic Manipulation, Autonomous Mobile Robotics, Embedded AI, Object-Oriented Programming & Design, Deep Learning, Human Robot Interaction, Mechanical Control Systems Design

Engineering Experience

R&D Embedded Firmware Intern

June 2025 – Sept 2025

Yaskawa

San Francisco, CA

- Revamped and refactored C++ driver firmware for prototype controller devices, restoring functionality for latch and coincident detection feature
- Designed and assembled hardware testing panels for controller and I/O verification, handling schematic design, wiring diagrams, component selection, and assembly to support in-house R&D validation
- Collaborated with firmware, hardware, and automation teams to troubleshoot multi-system integration issues

Embedded Systems Lead

Sept 2024 – Present

Northrup Grumman Collaboration Project (NGCP)

- Managed software and hardware sub-teams for development of an autonomous drone under a strict deadline
- Researched and integrated avionics components, sensors, and power systems into a custom UAV architecture to ensure electrical and mechanical compatibility
- Configured flight system, tuning control parameters, ESCs, and telemetry for stable multi-rotor operation

Projects

Pick and Place Robotic Arm | Python, Intel RealSense

Sept 2024 – Dec 2024

- Developed and implemented forward/inverse kinematics for precise control of the OpenManipulator-X robot arm
- Integrated visual servoing techniques and camera calibration to enhance feedback control, allowing real-time adjustments for improved grasp accuracy and stability
- Programmed various trajectory planning methods to enable the robot to reach target positions with high precision and smooth trajectory execution

TurtleBot | Python, ROS2, Jeston Orin Nano

Jan 2023 – Mar 2023

- Deployed autonomous navigation using ROS2 by interfacing with LIDAR, Jeston Nano, and an Oak-D Pro Camera
- Simulated navigation and mapping tasks in Gazebo, visualizing sensor data and robot behavior in RViz
- Configured and tested LIDAR-based room mapping using SLAM for environmental representation, implementing Adaptive Monte Carlo Localization (AMCL)

Imitation Learning for Mobile Robotics | Python, ROS2, Turtlebot4

Jan 2025 – Jun 2025

- Implemented LIDAR hallucination techniques to generate synthetic sensor data from simulation to create a large expert dataset
- Collecting and post processed navigation data from Gazebo and RVIZ simulation software
- Designed visualization tools to analyze sensor readings and navigation trajectories using Matplotlib and Pandas

RC Car with RTOS | C, STM32CubeIDE, STM Nucleo board, FreeRTOS

Mar 2024 – Jun 2024

- Developed an RC car using STM32 Nucleo boards in C, with FreeRTOS for real time Task management
- Designed custom wireless communication protocols for a receiver and transmitter for remote control functionality
- Implemented a low power mode to reduce power usage when the car was idle